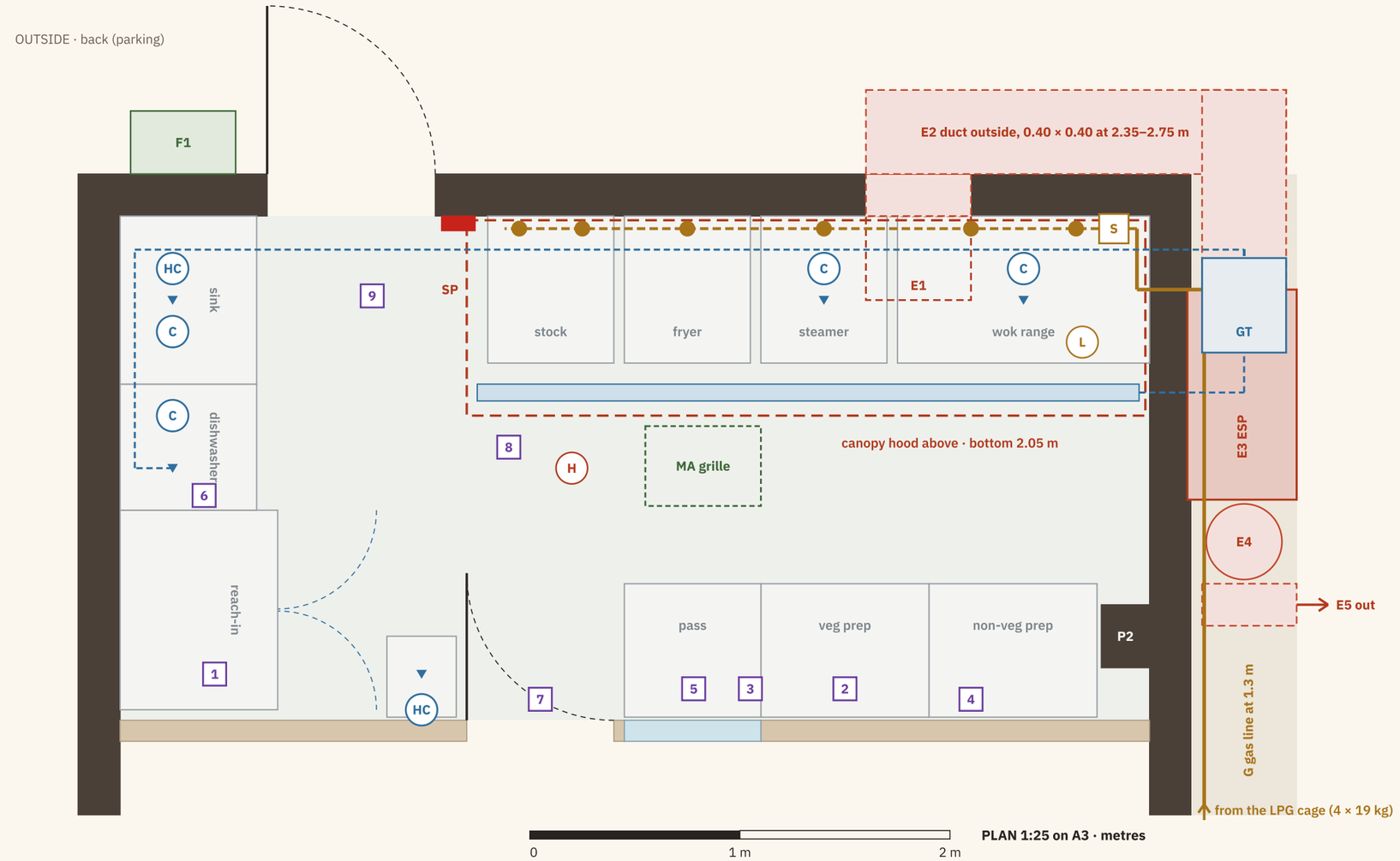
Kitchen services and exhaust

For kitchen equipment, exhaust, gas, plumbing and electrical vendors · plan at 1:25 and elevation at 1:40 on A3

Betalbatim pub · Simplex · Concept 27

Sheet 2 of 2 · A3

singhgenics.github.io/Simplex/kitchen.html



Services list

TAG	ITEM	SIZE / NOTE
E1	Duct through the back wall	0.40 × 0.40 at 2.35–2.75 m
E2	Duct along the wall, round the corner	0.40 × 0.40, outside
E3	ESP + carbon odour filter	0.50 × 1.00 × 0.90, 2.05–2.95 m
E4	Inline exhaust fan	Ø 0.44 · on backup
E5	Side discharge, mesh + rain cowl	0.40 × 0.40 at ~2.6 m
F1	Fresh-air intake fan, back wall	0.50 × 0.48 at 3.3 m
MA	Make-up air grille, ceiling	0.55 × 0.38
G	LPG line, copper, from the cage	outside wall at 1.3 m
S	Solenoid valve on the gas entry	interlocked: fan + suppression
SP	Hood suppression pull + gas cut-off	at the rear exit, 1.3–1.5 m
GT	Grease trap, outside back corner	0.45 × 0.45
1	Reach-in fridge	1-ph socket · on backup
2	Under-counter freezer	1-ph socket · on backup
3	Rice cooker	1-ph socket
4	Chilled mise-en-place rail	1-ph socket · on backup
5	Heat lamps over the pass	1-ph, switched
6	Dishwasher	1-ph or 3-ph: confirm model
7	Insect killer	1-ph
8	Hood lights	1-ph, switched
9	Exhaust fan + ESP control panel	by the rear exit · on backup

EXHAUST

Allow about 4,000–5,000 m³/h for the whole hood (vendor to size); about 8 m/s in the 0.40 × 0.40 duct. ESP + carbon filter before the fan. No roof stack: apartment above, parking behind. Make-up air about 80% of the exhaust, interlocked.

GAS

4 × 19 kg LPG in a locked, ventilated cage on the right strip. A valve per appliance; solenoid shuts off with the fan or the hood suppression; leak detector at the line entry.

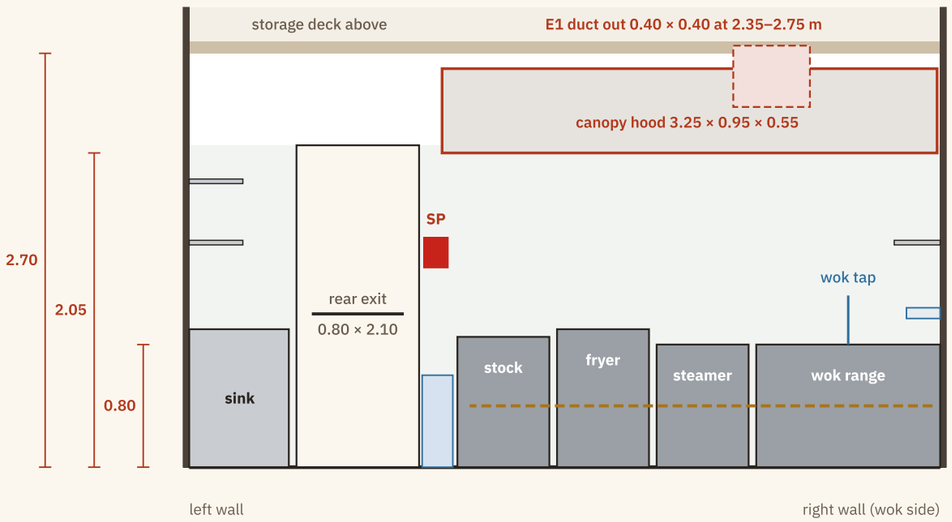
WATER + DRAINS

All kitchen drains to the grease trap, then the grey-water line (never the biodigester). Epoxy floor falls to the channel drain.

POWER

Shop supply: three-phase, about 15–20 kW. Backup 7.5 kVA covers the fridges, rails, fan, ESP, lights. Not the ACs.

Hot line elevation · back wall seen from inside · 1:40 on A3



Legend

- canopy hood outline (above)
- exhaust duct, outside, at 2.35–2.75 m
- gas line, copper
- gas manifold at 0.40 m · dot = appliance valve
- S — solenoid shut-off valve
- L — gas leak detector
- HC — water point: hot + cold (C = cold only)
- ▼ — drain point
- drain line to the grease trap
- 1 — power point, see the services list
- make-up (fresh) air grille in the ceiling
- H — heat detector
- SP suppression pull + gas cut-off